UNDERSTANDING NEWTON'S LAWS OF MOTION

THREE STRAIGHTFORWARD RULES DESCRIBE AND PREDICT HOW THINGS MOVE

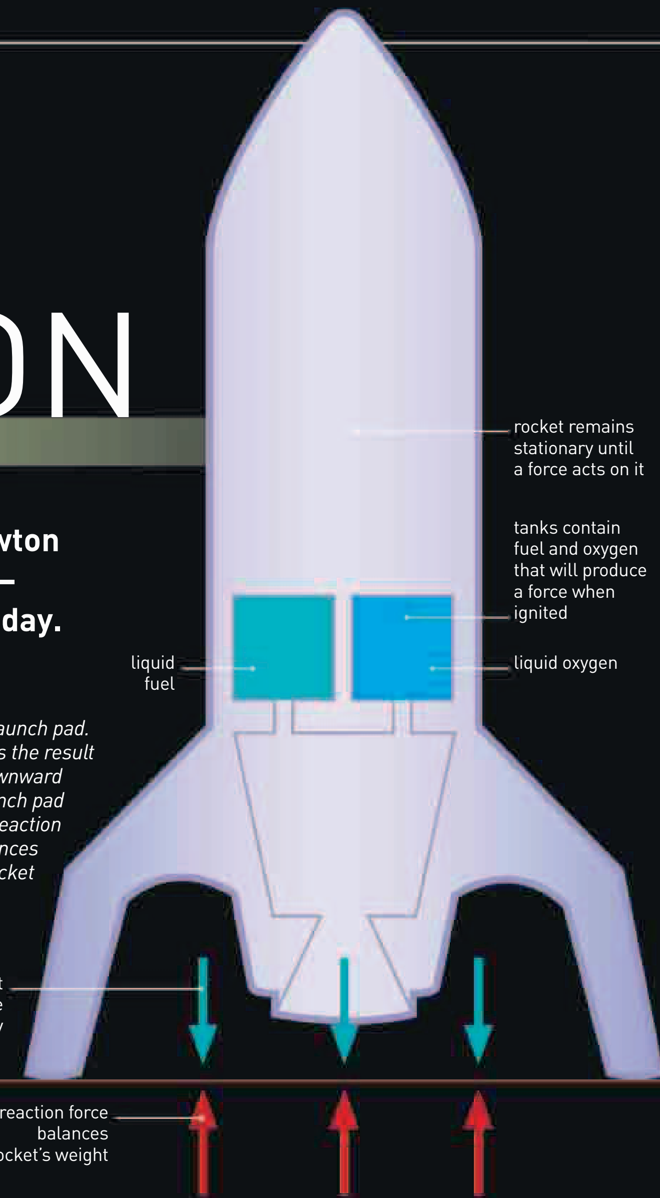
In the late 17th century, English physicist and mathematician Isaac Newton established the science of mechanics—the study of forces and motion—with three simple but revolutionary scientific laws that are still used today.

When Newton was a student, scholarly understanding of forces and motion was based on the ideas of ancient Greek philosopher Aristotle (384–322 BCE), who believed that an object moves only as long as a force acts on it. For example, according to Aristotle, projectiles in free motion are pushed along by following air currents. Thinkers in the Middle Ages expanded on this idea with the "impetus" theory that suggests that the force with which an object is thrown is stored in the object, and gradually runs out. Italian mathematician and physicist Galileo Galilei (1564–1642) overturned these ideas, realizing that an object continues to move at the same speed and in the same direction unless a force—such as gravity or air resistance—acts upon it. Newton adopted this idea as the first of his three laws, which he expressed in mathematical form in his book *Philosophiae Naturalis Principia Mathematica* (1687). Newton's laws accurately describe and predict the motion of objects in most situations. At very high speeds or in strong gravitational fields, they are not accurate because of effects explained by Einstein's theories of relativity (see pp.244–45).

ISAAC NEWTON
Newton was the most influential thinker and experimentalist of the 17th and 18th centuries. He made enormous contributions to the study of gravity, light, astronomy, and mathematics.

31,000

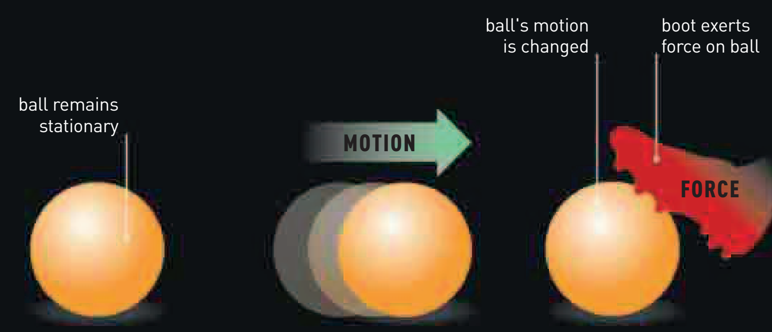
THE SPEED, IN MILES PER HOUR, AT WHICH THE VOYAGER 1 SPACECRAFT IS LEAVING THE SOLAR SYSTEM. **VOYAGER** KEEPS **MOVING** THROUGH SPACE BECAUSE **NO AIR RESISTANCE** ACTS ON IT.



ROCKET AT REST
A rocket stands on a launch pad. Its enormous weight is the result of gravity pulling it downward toward Earth. The launch pad produces an upward reaction force that exactly balances the weight, and the rocket does not move.

FIRST LAW

Newton's first law states that an object remains at rest or continues moving in a straight line unless a force acts upon it. Most objects have many different forces acting on them at all times, but often the forces balance. A book lying on a table, for example, is being pulled downward by gravity—but the table pushes upward on the book with a force of exactly the same magnitude (see third law). Since the forces balance, the book remains stationary.



AT REST
A ball remains stationary until a force acts upon it. The ball's weight pulls it downward, but the ground exerts an upward reaction force of the same magnitude, so the net force is zero.

IN MOTION
Once the ball is in motion, its velocity—or particular combination of speed and direction—continues. In reality, friction between the ball and the surface would slow it down.

FORCE APPLIED
A force, such as a kick from a boot, alters the ball's velocity, a change termed acceleration. The ball either slows down, speeds up, or changes its direction with or without changing its speed.